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Mechanical Properties of Graphite Filled Unsaturated Polyester and Unsaturated Polyester/Palm Oil Blend Resin

Siti Noor Hidayah Binti Mustapha[1,a*],
Muhammad Syafiq Asyraf Bin Azhman[2,b], Shamsul Zakaria[3,c],
Rasidi Roslan[4,d], Rohani Binti Mustapha[5,e] and Lee Tee Chuan[6,f]

[1,2,3,4]Faculty of Industrial Sciences & Technology, Universiti Malaysia Pahang, Lebuhraya Tun Razak, 26300 Gambang, Pahang, Malaysia

[5]School of Ocean Engineering, Universiti Malaysia Terengganu, 21030, Kuala Terengganu, Terengganu, Malaysia

[6]Faculty of Technology Management and Business, Universiti Tun Hussein Onn Malaysia, 86400, Batu Pahat, Johor

[a]snhidayah@ump.edu.my*, [b]asyrafazhman14@gmail.com, [c]shamsulzakaria@ump.edu.my, [d]rasidi@ump.edu.my, [e]rohani.m@umt.edu.my, [f]tclee@uthm.edu.my

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Abstract. This research is aimed to investigate the effect of graphite loadings in unsaturated polyester (UPE)/acrylated epoxidized palm oil (AEPO) blend resin. The modification of epoxidized palm oil (EPO) to AEPO was carried out using acrylation process and further blended with synthetic UPE resin. Graphite powder was added at 0.03, 0.05 and 0.1 phr into the neat UPE and UPE/AEPO blend resin further cured in an oven at 100 °C and 160 °C. FTIR spectrums showed the disappearance of oxirane ring and existence of carbon double bond indicating successful of AEPO synthesis process. Tensile and Izod impact test revealed that graphite showed different effects to neat UPE and UPE/AEPO blend resin. In neat UPE, graphite significantly improved the stiffness properties at 0.1 phr additions. However, in UPE/AEPO blend resin, the toughness properties were improved with increased graphite loadings.

Introduction

Replacing petroleum-based polymer materials with material derived from renewable re-sources has attracted a great deal of attention in recent years due to their importance in environment and sustainability point of view. In the last decade, vegetable oil has been intensively explored to replace the synthetic thermoset resin such as polyester in the market. The most common vegetable oil used are linseed oil, soybean oil, jatropha oil and many more [1-3].

Palm oil is the world's highest yielding oil crop with an output of 5–10 times greater per hectare than other vegetable oil [4]. It is abundantly available in Malaysia and Indonesia [5]. Previous studies have proved that palm oil has lower mechanical and thermal properties that are not suitable to be used to replace polyester resin in the market [6]. This is due to lower unsaturation in the triglyceride structures of palm oil that was indicated by the lower iodine value [6]. In order to overcome this issue, the palm oil structures need to be modified chemically allowing higher unsaturation to initiate polymerization process [7]. The most common modification methods are epoxidation and acrylation [8, 9]. In addition, blending some part of the polyester with modified palm oil has provided a promising alternative to deal with the sustainable and environmental issues without compromising its mechanical and thermal performances [6, 10]. Besides the modification of the triglyceride structure, addition of filler is one of the approaches to improve the properties of vegetable oil based resin [7, 11, 12]. The most common filler used is montmorillonite nanoclay [13-15] and functionalized graphene [16, 17].

Graphite is a crystalline form of carbon element arranged in hexagonal structure, which is essentially made up of hundreds to thousands layers of graphene [18]. It can be found naturally or can be produced synthetically. Graphite is a very conductive materials with an electrical conductivity of 10^4 S/cm at ambient temperature [19]. Graphite is also known as the hardest material as it is stronger than diamond when in plane and has high stiffness with the value of Young's modulus of around 1060 MPa, which is several times greater than clay [20]. Graphite also possesses excellent thermal and electrical conductivity. Therefore, due to its outstanding properties, graphite has been frequently used as reinforcement filler in polymer material [21]. This research aims to study the mechanical effect of graphite filler in UPE blended with acrylated modified palm oil resin. As a control, graphite filled neat UPE resin will also be evaluated.

Materials

The main reagents and materials used in the analysis were graphite powder, PN 104206, MerckMillipore. The UPE graded as Reversol P-9539 was purchased from Impian Z Sdn. Bhd. Epoxidised palm oil (EPO) was received from Budi Oil Sdn. Bhd. Acrylic acid was purchased from Merck, Germany. Triethylamine, supplied by Aldrich, UK. Hydroquinone was purchased from Nacalai Tesque, Japan. Benzoyl peroxide was purchased from Merck, Germany. Diethyl ether was received from R&M Chemicals.

Synthesis of Acrylated Epoxidized Palm Oil (AEPO) Resin

Acrylic acid was mixed with EPO at ratio of 1:4 (w/w). Hydroquinone and triethylamine were added to the mixture as inhibitor and catalyst, respectively. The mixture was then stirred using mechanical stirrer with temperature set at 80 ºC and rotation speed of 300 rpm. The AEPO mixture was then washed with petroleum ether and distilled water before the mixture was further dried using rotary evaporator.

Preparation of Unsaturated Polyester (UPE)/Acrylated Epoxidized Palm Oil (AEPO)/Graphite Blend Resin

Graphite powder was immersed in acetone solution and sonicated for 90 minutes. AEPO and UPE resin were then being added at ratio 10:90 (w/w) to the graphite solution and mixed thoroughly using homogenizer for 24 h at 60 ºC. Benzoyl peroxide was added at 1.5 phr and cured at 100 °C for 2 h and post cured at 160 °C for another 2 h in an oven.

Characterization

Fourier Transform Infrared Spectroscopy (FTIR). FTIR test was conducted using Perkin Elmer 1600 FTIR spectrophotometer. Attenuated total reflectance (ATR) method was used in this study. The samples was dropped onto the sample disc holder and scanned 16 times. The FTIR spectra was recorded for the range of 400 to 4000 cm^{-1}.

Tensile Test. Instron Universal Testing Machine was used to analyze the tensile properties of graphite filled in UPE and UPE/AEPO blend resin. The samples were cut into dumbbell shape referring to ASTM D638 (type V) standard. Load was applied at constant crosshead speed of 0.5 mm/min at room temperature. Elongation at break, tensile modulus, and tensile strength were evaluated from the stress-strain data. Six specimens were tested and the mean of the impact resistance was determined.

Izod impact test. Impact strength of the samples was measured using Izod impact tester (Zwick, Germany) according to the ASTM D256 standard. Specimens with dimensions of 13 mm width, 54 mm length and 3 mm thickness were used with notched depth and radius of 1.5 mm and 0.25 mm, respectively.

Result and Discussions

Fig. 1 shows the FTIR spectra of EPO and the synthesized AEPO. In EPO spectrum, small peaks at 840 cm^{-1} (oxirane group, COC) and 1240 cm^{-1} (C-O) were seen indicating the existence of epoxy group. Both of the peaks were observed to be small due to the low existence of epoxy group in the palm oil triglycerides. Besides that, additional characteristics of the epoxy were noted at peak 2924 cm^{-1}, 2825 cm^{-1} and 1378 cm^{-1}, indicating the existence of the C-H stretching of terminal methyl (CH_3) and methylene (CH_2) [8]. Strong carbonyl stretching (C=O) was also clearly seen at 1741 cm^{-1}.

Meanwhile, for the AEPO, peak at 840 cm^{-1} (oxirane group, COC) had disappeared and broad peaks was observed at 3524 cm^{-1} indicating the oxirane ring was successfully opened and the acrylation process had successfully occurred [9, 22]. The existence of new peak at 1635 cm^{-1} (C=C) also confirmed the acrylation process had successfully occurred [7]. The COC ring opened and allowed OH group to be occurred at one chain and acrylation C=C at another chain.

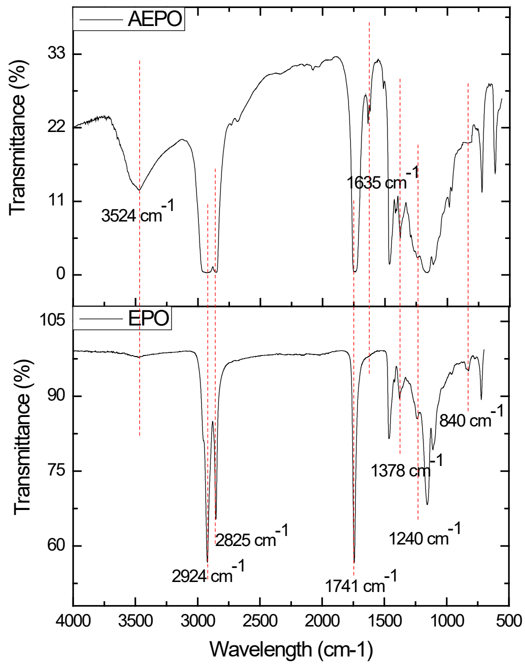

Fig. 1. FTIR spectra of epoxidized palm oil (EPO) and acrylated epoxidized palm oil (AEPO).

Fig. 2 shows the tensile modulus of UPE and UPE/AEPO blend resin with the addition of 0, 0.03, 0.05 and 0.1 phr of graphite powder. As can be seen from Fig. 2, the addition of low amount of graphite, which is 0.03 and 0.05 phr, decreased the tensile modulus properties of the neat UPE as the Young's modulus value dropped from 279.1 MPa to 13.61 MPa and 51.43 MPa, respectively. This is owing to the poor graphite distribution and interaction in the UPE resin matrix, as the improvement of the mechanical properties of graphene composites relies on the interface interaction between graphite and the polymer matrix [17]. In addition, previous research has proven that the low amount of graphite filler had not significantly affected the tensile modulus improvements in thermoset resin [19]. However, at higher graphite loadings (0.1 phr), the increase in Young's modulus of UPE resin to 1233.2 MPa contributed to 349.4% improvement in stiffness properties of UPE resin. This is mainly attributed to the higher interactions of graphite with UPE resin at higher loadings. The high stiffness properties of graphite in nature, which consist of strong covalent bonding of carbon crystalline structure, contributed to the higher modulus properties.

In UPE/AEPO blend resin, the addition of graphite decreased the Young's modulus properties of the resin. The Young's modulus values were dropped from 322.8 MPa to 81.09 MPa, 68.73 MPa and 49.45 MPa at 0.03, 0.05 and 0.1 phr graphite loadings, respectively. This phenomenon could be due to poor distribution of graphite in the UPE/AEPO resin system, which could not provide good interactions between the resin and graphite system. Thus, stress could not be transferred accordingly.

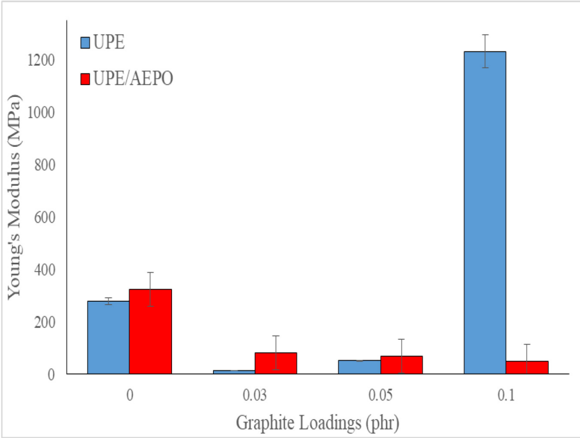

Fig. 2. Tensile modulus of UPE/Graphite and UPE/AEPO/Graphite Resin Blend.

Fig. 3 and 4 show the elongation at break and Izod impact strength of UPE at UPE/EPO resin at the addition of 0.03, 0.05 and 0.1 phr graphite. As can be seen from the graphs, the addition of 0.03 phr graphite in UPE increased the elongation at break and impact strength from 3.04% to 9.83% and 25.28 kJ/m^2 to 75.20 kJ/m^2, respectively. These results can be attributed to the lower stiffness properties as shown in Young's modulus value. Thus, it was revealed that the addition of 0.03 phr graphite improved the flexibility and toughness properties of UPE resin. However, at higher graphite loadings, which were 0.05 and 0.1 phr, a decrement in elongation at break and impact strength properties was shown. This is due to the lower toughness properties, as graphite has hindered the movement of polymer chains; thus, the composites are unable to dissipate the external mechanical energy.

In UPE/AEPO resin, the addition of graphite increased the elongation at break and impact strength properties. As can be seen in Figure 3 and 4, the elongation at break value increasing from 1.82% to 2.14%, 2.85% and 4.58% while the Izod impact strength from 1.90 kJ/m2 to 6.30, 7.76 and 9.48 kJ/m^2 at 0.03, 0.05 and 0.1 phr graphite additions, respectively. This phenomenon proved that the additions of graphite significantly increased the flexibility and toughness properties of the UPE/AEPO resin system. It could be explained by the existence an intermediate medium in the system provided by graphite, which allowed system to elongate and absorb higher energy before fracture.

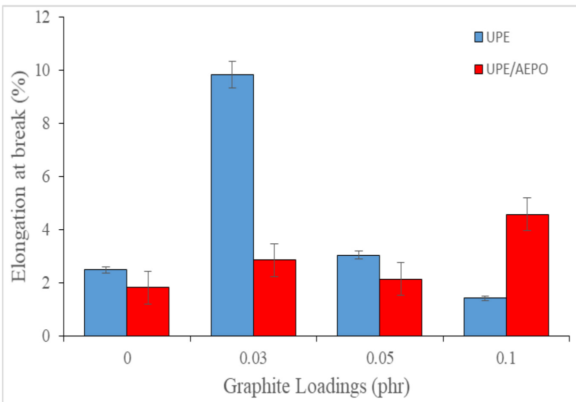

Fig. 3. Elongation at break (%) of UPE/Graphite and UPE/AEPO/Graphite Resin Blend

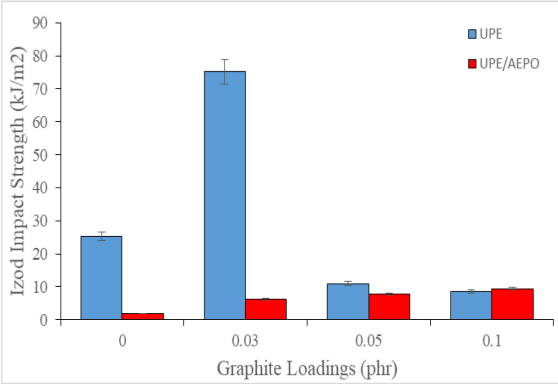

Fig. 4. Impact strength (kJ/m^2) UPE/Graphite and UPE/AEPO/Graphite Resin Blend

Conclusion

The AEPO is successfully synthesized from EPO through acrylation process. The ring of epoxy is successfully opened and acrylation process is evident by the existence of carbon double bond and hydroxyl peaks in FTIR spectra. In this study, graphite is proven to have different effects to neat UPE and UPE/AEPO blend resin at different loadings from 0.03 up to 0.1 phr. In short, graphite is proven to have significant effect at 0.01 phr as it contributed to higher stiffness properties. However, in UPE/AEPO blend resin, graphite additions apparently contributed to higher elongation and toughness properties of resulted resin. This phenomenon proved that the addition of graphite provided an intermediate medium to UPE/AEPO blend resin, which increased its flexibility and allowed higher energy to be absorbed prior to application of sudden impact forces.

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References

[1] W. Liu, T. Xie, and R. Qiu, Biobased thermosets prepared from rigid isosorbide and flexible soybean oil derivatives, ACS Sust. Chem. & Eng., 5(1) (2016) 774-783.

[2] M.D. Samper, et al., Properties of biobased epoxy resins from epoxidized linseed oil (ELO) crosslinked with a mixture of cyclic anhydride and maleinized linseed oil. eXPRESS Poly. Lett. 13(5) (2019) 407-418.

[3] F.E.A. Latif, et al., Physicochemical, Structural And Mechanical Evaluation Of Bio-Based Epoxidized Jatropha Oil Blended With Amine-Cured Epoxy Resin As New Hybrid Matrix. J. of Eng. Sci. and Tech. 14(2) (2019) 1071-1087.

[4] L.S. Woittiez, et al., Yield gaps in oil palm: A quantitative review of contributing factors. Eur. J. of Agro. 83 (2017) 57-77.

[5] R. Abd Majid, Regenerated Spent Bleaching Earth for the Decolourisation and Bod Reduction of Palm Oil Mill Effluent. J. of Oil Palm Res., 2017.

[6] S. Mustapha, et al., Novel bio-based resins from blends of functionalised palm oil and unsaturated polyester resin. Mat. Res. Inno. 18 (2014) S6-326-S6-330.

[7] S. Mustapha, A.R. Rahmat, and A. Arsad, Bio-based thermoset nanocomposite derived from vegetable oil: a short review. Rev. in Chem. Eng. 30(2) (2014) 167-182.

[8] A. Salih, et al., Synthesis of radiation curable palm oil–based epoxy acrylate: NMR and FTIR spectroscopic investigations. Mol 20(8) (2015) 14191-14211.

[9] A. Fakhari, et al. Synthesis of new bio-based thermoset resin from palm oil. in Advanced Materials Research. Trans Tech Publ. (2014).

[10] S.G. Tan and W.S. Chow, Thermal Properties, Fracture Toughness and Water Absorption of Epoxy-Palm Oil Blends. Poly-Plastics Tech. and Eng. 49(9) (2010) 900-907.

[11] R. Mustapha, et al., Vegetable oil-based epoxy resins and their composites with bio-based hardener: a short. (2019).

[12] M.A. Mosiewicki and M.I. Aranguren, A short review on novel biocomposites based on plant oil precursors. Eur. Poly. J. 49(6) (2013) 1243-1256.

[13] S.N.H. Mustapha, et al., Flexural Properties of MMT Reinforced Unsaturated Polyester/Epoxidized Palm Oil Biobased Resin. Adv. Mat. Res. 1112 (2015) 377-380.

[14] A. Fakhari, et al., Influence of Nanoclay on Mechanical Properties of Vinyl Ester/Acrylated Epoxidized Palm Oil Blends. App. Mech. and Mat. 695 (2014) 293-296.

[15] Z.S. Alsagayar, et al., Tensile and Flexural Properties of Montmorillonite Nanoclay Reinforced Epoxy Resin Composites. Adv. Mat. Res. 1112 (2015) 373-376.

[16] J. Zang, et al., Fracture Behaviors of TRGO-Filled Epoxy Nanocomposites with Different Dispersion/Interface Levels. Macromol. Mat. and Eng. 300(7) (2015) 737-749.

[17] B. Thalib Nur, et al., Tailoring graphene reinforced thermoset and biothermoset composites, in Rev. in Chem. Eng. (2019).

[18] C. Cong, et al., Self-limited oxidation: a route to form graphene layers from graphite by one-step heating. Small, 6(24) (2010) 2837-41.

[19] B. Debelak and K. Lafdi, Use of exfoliated graphite filler to enhance polymer physical properties. Carbon, 45(9) (2007) 1727-1734.

[20] A. Naz, A. Kausar, and M. Siddiq, Influence of graphite filler on physicochemical characteristics of polymer/graphite composites: a review. Poly-Plas Tech. and Eng. 55(6) (2016) 604-625.

[21] B. Suresha, G. Chandramohan, and N. Renukappa, Mechanical and tribological properties of glass–epoxy composites with and without graphite particulate filler. J. of app. poly. sci. 103(4) (2007) 2472-2480.

[22] D.P.P. Kosheela., A.H. Hazimah, and N.M.T.I. Tuan, Optimization on Synthesis of Acrylated Epoxidized Palm Olein using Response Surface Methodology. J. of Oil Palm Res. 27(4) (2015) 366-376.